1. A user is required to enter a valid number in a form, but users sometimes input invalid data.
   Write logic to repeatedly prompt the user until they enter a valid integer.

   LOGIC:

   1. Input valid number from the user.
   2. Check whether the number is valid or invalid.
   3. If the number is valid then execute otherwise ask the user for prompt, until the user get a valid integer.

2. A data analysis tool processes a list of numbers and needs to identify the most frequently occurring value.

   LOGIC:

   1. Import Statistics module which is built in Python.
   2. Get the list of numbers from the user.
   3. Use Syntax statistics.mode() to get the most frequently occurring value.
   4. When call the function it will return the most common element in the list.

3. A text-processing application needs to compare words and check if they are anagrams (contain the same letters in a different order).

   LOGIC:

   1. Get the two inputs string from the users.
   2. Convert both words to lower case without fail.
   3. If length of the words differs, they can’t be anagrams.
   4. If length of the words matches, sort the letters and compare.
   5. That words are anagrams.

4. A speech analysis program needs to count the number of vowel sounds in a given input.

   LOGIC:

   1. Read the input string.
   2. Define the set of vowels in caps and small (A,E,I,O,U and a,e,i,o,u)
   3. Initialize count to zero.
   4. Loop each character from the string.
   5. If the character vowel add to count. Return total count of vowel.

5. A text-editing software includes a feature to reverse the order of words in a sentence for stylistic effects.

LOGIC:

1. Read the input sentence.
2. Split sentence in words into the list.
3. Reverse the order of words in the list.
4. Join back into the string.
5. Call the reversed sentence.

6. A missing number is detected in a sequence of values stored in a database.

LOGIC:

1. Read the list of numbers.
2. Use the formula n * (n + 1) / 2 to calculate the sum of numbers from 1 to n.
3. Calculate the actual sum of numbers in the given list.
4. Subtract the actual sum from the expected sum to find the missing number.
5. Display the missing number.

7. An ATM machine processes withdrawal requests and needs to ensure that users cannot withdraw more than their account balance.

LOGIC:

1. Read the customer's account balance.
2. Get input from the user to withdraw the amount.
3. If the amount is lesser than or equal to the balance amount, proceed to withdraw.
4. If the amount is higher than the balance amount, display "Insufficient Balance".

8. A system needs to verify whether a given dataset contains duplicate entries.

LOGIC:

1. Read the data from the dataset.
2. Check if the length of data is equal or not equal to set data.
3. If the data length is equal to the length of set(data) print ”No Duplicates”.
4. If the data length is not equal to the length of set(data)print “Duplicates found”.

9. A digital calculator includes a feature to sum the digits of a number for verification purposes.

LOGIC:

1. Read the input number from the user.
2. Convert the number into individual digits.
3. Initialize the variable sum to zero.
4. Sum the number of digits through looping each number.
5. Return the sum of digits.

10. A language-learning app wants to verify whether a given sentence is a pangram (contains every letter of the alphabet at least once).

LOGIC:

1. Read the number of characters in the given string.
2. Convert the sentence to lowercase and remove the non alphabetic characters.